NATIONAL TEACHERS COLLEGE
School of Arts, Science and Technology
629 J. Nepomuceno, Quiapo, Manila

Smart Inventory/Waste Management System
SDG 12: Responsible Consumption

A Software Design and Analysis Document
Presented to the faculty of
School of Arts, Science and Technology of
National Teachers College

In Partial Fulfillment
of the Requirement for
IT Elective 2

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I. Introduction

1.1 Project Overview:

The **Smart Waste Management System** is a desktop-based application designed to help barangays efficiently record, monitor, and manage waste collection data. The system allows users to input waste entries, categorize them (e.g., plastic, paper, food), and generate reports for monitoring environmental impact.

The vision of the project is to **promote proper waste segregation, improve data tracking, and support environmentally sustainable practices** at the community level.

This project supports the **United Nations Sustainable Development Goals**, specifically:

- **Goal 11: Sustainable Cities and Communities** – improving waste management systems
- **Goal 12: Responsible Consumption and Production** – encouraging proper waste segregation and monitoring

1.2 Problem Statement: Rationale for the application's existence.

Many barangays still rely on **manual recording (paper-based systems)** for tracking waste collection. This leads to:

- Data loss or misplacement
- Inaccurate waste tracking
- Difficulty in generating reports
- Lack of proper monitoring of waste types

The Smart Waste Management System was developed to address the inefficiencies of manual waste tracking methods, which often result in delayed decision-making and difficulty implementing effective waste management programs. By digitizing the process through a centralized platform, the system enables faster data collection, more accurate record-keeping, and simplified management of waste information, ultimately supporting data-driven decisions for more effective waste reduction initiatives.

II. Requirements & Analysis

- **2.1 Functional Requirements:** (FR1, FR2, etc.)

ID	Requirement	Alignment
FR1	**User Authentication:** The system must allow users to log in securely using a username and password (admin/admin).	
FR2	**Add Waste Entry:** The system must allow users to add waste entries including Household Name, Waste Type, Waste Amount (kg), and Date.	
FR3	**Display Waste Records:** The system must display all waste entries in a DataGridView table for easy viewing.	
FR4	**Calculate Total Waste:** The system must compute and display the total waste collected in kilograms.	
FR5	**Total Records Count:** The system must show the total number of waste entries recorded.	
FR6	**Recent Entry Display:** The system must display the most recent waste entry (household and waste type).	
FR7	**Generate Reports:** The system must provide a report view showing all waste records with summary statistics.	

FR8	**Export Reports:** The system must allow users to download waste reports as CSV files.	
FR9	**Navigation:** The system must allow seamless navigation between Dashboard and Reports modules.	

- **2.2 Non-Functional Requirements:** Performance, Security, and Robustness metrics.

ID	Category	Requirement	Metric
NFR1	Performance	Data Loading	The system should load waste records within 1–2 seconds.
NFR2	Performance	Report Generation	Report generation should be completed instantly for small datasets (< 50 records).
NFR3	Performance	Total Waste Calculation	Real-time calculation upon adding new entries.
NFR4	Security	User Authentication	Login system with predefined credentials (admin/admin).
NFR5	Security	Input Validation	Prevent invalid data entry (empty fields, non-numeric waste amounts).
NFR6	Robustness	Null-Handling	System should handle null values gracefully without crashing.

NFR7	Robustness	Error Handling	Database connection errors should be caught and displayed properly.
NFR8	Robustness	Crash Prevention	System should not crash during invalid input; display appropriate error messages instead.

- **2.3 Complexity Analysis:** Big O Notation analysis for the core Business Logic method (e.g., the algorithm in the Service Layer).

Method/Operation	Algorithm	Time Complexity	Justification (Big O Notation)
Load Waste Records	SELECT query + DataTable Fill	O(n)	Retrieves all records from database; time grows linearly with number of records.
Calculate Total Waste	SUM aggregation	O(n)	Iterates through all records to compute total (or uses SQL SUM).
Add Waste Entry	INSERT query	O(1)	Single insert operation, constant time regardless of dataset size.
Display Recent Entry	Last row access	O(1)	Direct access to the most recent record.
CSV Export	Iterate through DataGridView	O(n)	Loops through all rows to write to file.
Linear Search in Login	Simple string comparison	O(1)	Fixed credential check (admin/admin).

III. Design Specification

- **3.1 N-Tier Architecture Diagram:** Visual representation of the layer interactions.

  The system follows a classic 3-Tier Architecture which separates concerns into three distinct layers: Presentation Layer, Business Logic Layer, and Data Access Layer. This separation ensures modularity, maintainability, and scalability of the application. There has been a problem in our device which is unable to separate the BLL, DAL and many more which forces use to use an SQL lite bench to store the data that is submitted in the application. But we still managed to give the result by analyzing the code and connecting it into the required part in the github as in the following below.

Layer	Components	Responsibilities
Presentation Layer (UI)	• LoginForm.cs • Form1.cs (Dashboard) • ReportForm.cs	• User interface and interaction • Data input and display • Navigation between modules • User input validation
Business Logic Layer (BLL)	• Total waste calculation • Record count aggregation • Recent entry detection • Report summary generation	• Core business rules and computations • Data processing before display • Coordinates between UI and Data layers
Data Access Layer (DAL)	• DatabaseHelper.cs • SQLite connection management • CRUD operations	• Encapsulates database operations • Executes SQL queries • Manages database connections • Handles data retrieval and persistence
Database Layer	• SmartWaste.db (SQLite) • WasteEntries table	• Physical data storage • Data integrity and persistence • SQL query execution

- **3.2 Entity Relationship Diagram (ERD):** Detailed database schema including data types and Foreign Key relationships.

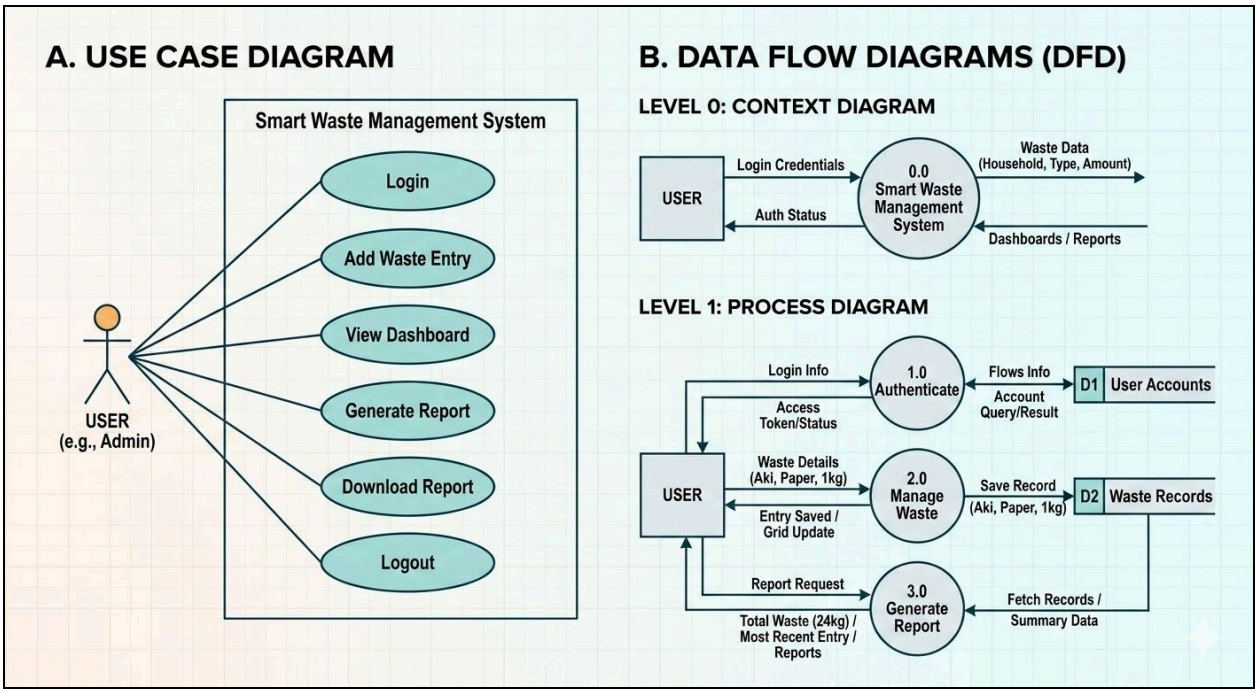
1. Users Table This table handles the login credentials

Field name	Data type	Description
UserID	INT	Unique ID for each system user
Username	Varchar(50)	User login name (“admin”)
Password	Varchar(255)	Encrypted string for authentication
Role	Varchar(20)	Defines access levels (Admin/staff)

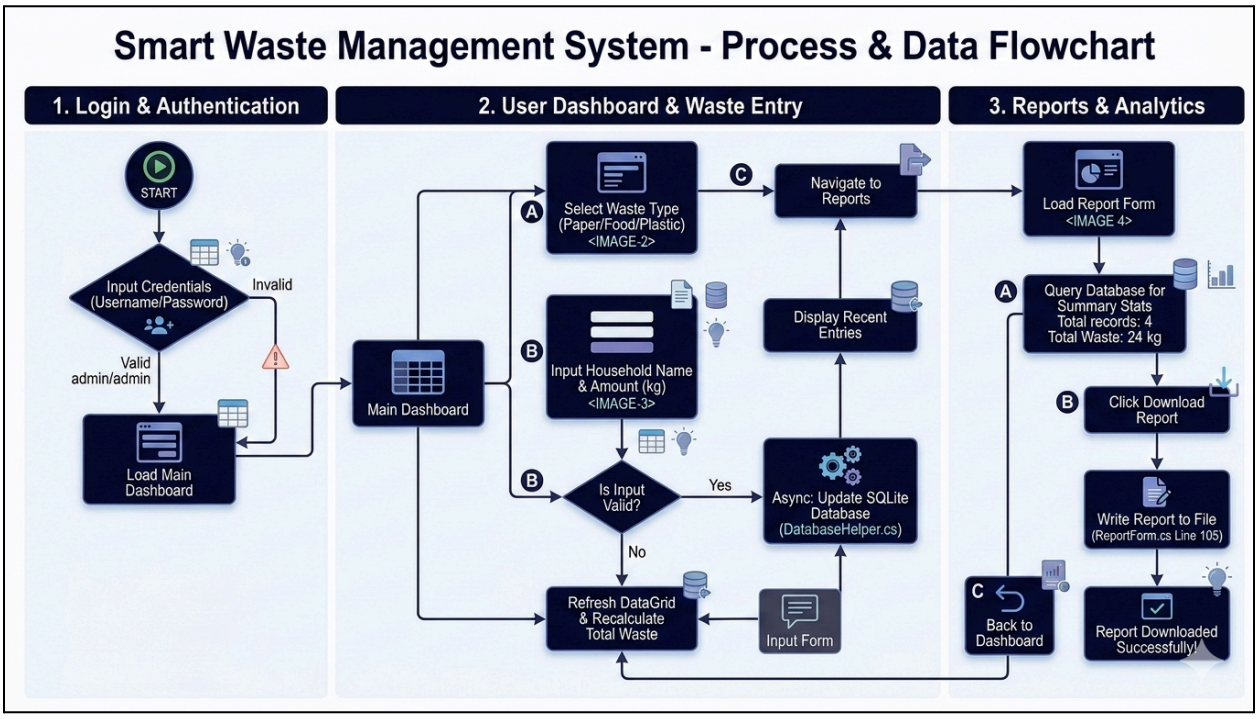
2. WasteRecords Table This table stores the entries made through the "Add Waste Entry" dashboard.

Field name	Data type	Decription
RecordID	INT	Unique ID for each waste entry
HouseholdName	VARCHAR(100)	Name of the source
Wastetype	VARCHAR(50)	Category (Paper,Plastic,Food)
WasteAmount	FLOAT	Weight in kg
EntryDate	DATE	Date selected from UI picker
AddedBy	INT	Reference

- **3.3 Use Case & Data Flow Diagrams (DFD):** Modeling user interactions and data movement.Level 0 and Level 1 for DFD



- **3.4 Algorithm Flowchart:** Step-by-step logic of the Async or Business method.



IV. Individual Contributions

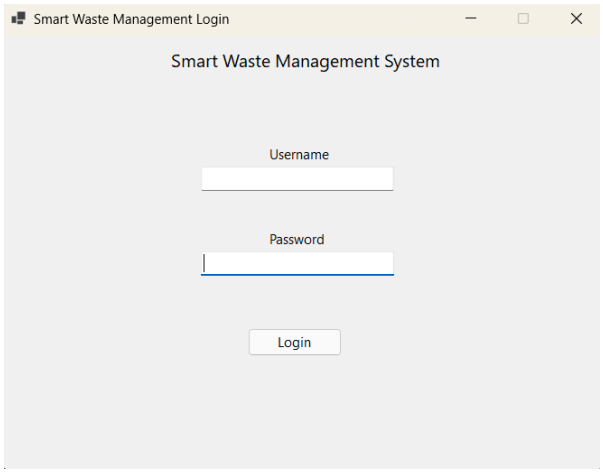
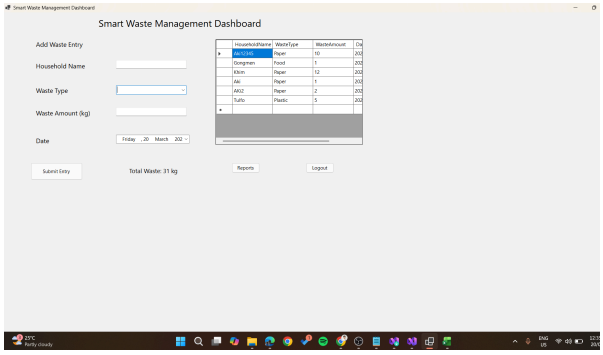
- A table detailing exactly which Layer or Module was implemented by which member.

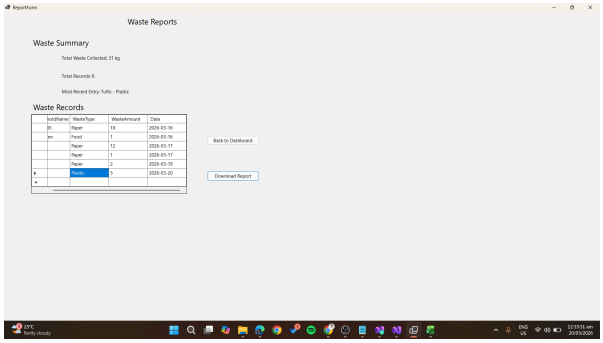
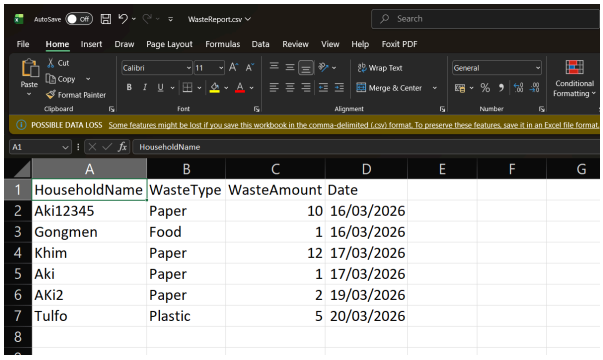
TASKS	MEMBERS
Application in Visual Studio	**Developer:** Sabio, Khim Axelle **Recommendations and Improvements:** Nicole Manalili Shin Adrian Gamao Marvin Quilang
Documentation and PPT	**Documentator and PPT:** Nicole Manalili Shin Adrian Gamao **Assistance:** Marvin Quilang **Revisions and Checking:** Sabio, Khim Axelle
Diagrams and Structures	**Creator:** Marvin Quilang **Recommendations:** Shin Adrian Gamao Nicole Manalili **Revisions and Checking:** Sabio, Khim Axelle
Github Repository Creation	**Creator:** Nicole Manalili **Recommendations:** Shin Adrian Gamao Marvin Quilang Sabio, Khim Axelle

Overall Note: All of the members have contributed to the creation of the Final Project at IT Elective 2, guiding one another and spotting each other on what to change and what to do.

V. Appendices

- Screenshots and description of the application

Login Page	
Dashboard	

Waste Report	
CSV Result	

DESCRIPTIONS:

- **Login Page:**
  First is the user must log in into the admin using their credentials, in this case the username and the password in both “ admin “

- **Dashboard:**
  After Logging in, there will be a dashboard where it will show you where to put all of the informations and after that you will submit it and in the data grid in the left side it will show the past inserts of information

- **Waste Report:**
  After the report was finished it will also go to the waste report where it will show you the total among of waste, what is the most recent entry and

the data collected there are also an option to download the whole data

- **CSV Result:**
  After you click download it will be saved in the local file of your laptop and it will be saved there in CSV Format/